Derivations for FliBe/pebbles/TRISO particles

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1. Pebble power and fluid-pebble interface

- Cell power density $\acute{q}_c$ to:
  - Representative pebble (subscript p) power and matrix density (all cell power from pebbles):

$$\acute{Q}_p = \acute{q}_c \times V_c$$

$$\acute{q}_p = \frac{\acute{Q}_p}{V_p} = \frac{\acute{Q}_p}{\frac{4}{3}\pi R_p^3}$$

  - Fuel kernel (subscript f) power and density (all pebble power from kernels in the N_T TRISO particles). TRISO with subscript T:

$$\acute{Q}_F = \acute{Q}_T = \frac{\acute{Q}_p}{N_T}$$

$$\acute{q}_f = \frac{\acute{Q}_f}{V_f} = \frac{\acute{Q}_f}{\frac{4}{3}\pi R_f^3}$$

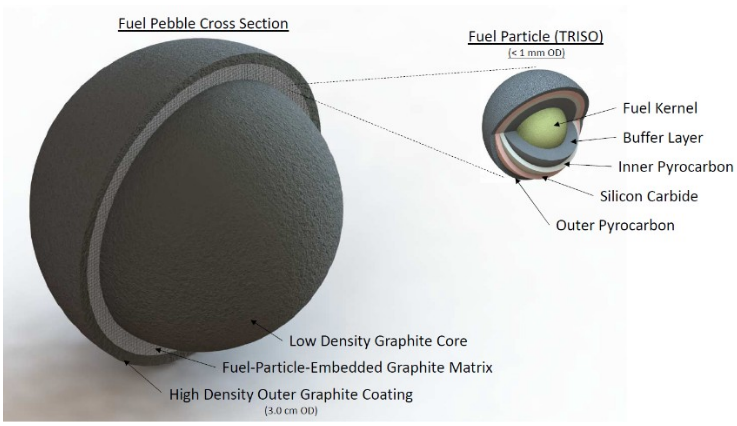
- Convection boundary condition between fluid (subscript F) T_F and representative pebble surface $T_{p,S}$ (for coupling). Pebble outer surface is interface surface:

$$q''_{p\to F} = \frac{\acute{Q}_p}{S_p} = \frac{\acute{Q}_p}{4\pi R_p^2}$$

$$q''_{p\to F} = h\left(T_F - T_{p,S}\right) \to T_{p,S} = \frac{\acute{q}''}{h} + T_F = \frac{q''_{p\to F} + hT_F}{h}$$

2. Pebble

- Three zones in the pebble: core, matrix, shell (subscripts c, m, s, respectively).
- R represents outer radius of each zone.
- 1D conduction in each layer, power generation only in the matrix.
- More advanced model (not covered) with k, N, BU, T dependence: https://inldigitallibrary.inl.gov/sites/sti/sti/5872007.pdf



a) Shell

- Pebble surface temperature is outer shell temperature:

$$T_s\left(R_s\right) = T_{p,S}$$

- Profile temperature in shell with conductivity of graphite k_g:

$$T_s(r)=T_{p,S}+\frac{\acute{Q}_p}{4\pi k_g}\times\left(\frac{1}{r}-\frac{1}{R_s}\right), r\in[R_m,R_s]$$

This can be rewritten as:

$$\acute{Q}_p=H_{shell}(T_m-T_s)$$

where

$$H_{shell}=\left(\frac{1}{4\pi k_g}\times\left(\frac{1}{R_m}-\frac{1}{R_s}\right)\right)^{-1}$$

is the conductance associated with the shell.

b) Matrix

- Matrix power density:

$$\acute{q}_m=\frac{\acute{Q}_p}{V_p}=\frac{\acute{Q}_p}{\frac{4}{3}\pi\left(R_m^3-R_c^3\right)}$$

Profile

- Derivation for temperature profile (for k_{eff}, see below):

$$\begin{cases} k_{eff}\Delta T_m=\acute{q}_m \\ \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dT_m}{dr}\right)=\frac{\acute{q}_m}{k_{eff}} \end{cases}$$

- Solving with flux $\psi=\frac{dT}{dr}$, the solution is of the form:

$$\psi=\frac{-r\acute{q}_m}{3k_{eff}}+\frac{C_1}{r^2}$$

- **Boundary condition 1**:

$$T_m(R_m)=T_s(R_m)=T_{p,S}+\frac{\acute{Q}_p}{4\pi k_g}\times\left(\frac{1}{R_m}-\frac{1}{R_s}\right)$$

- **Case 1**: $R_c=0$ (no graphite core)
  Boundary condition 2 is zero flux:

$$\frac{dT_m}{dr}(0)=0$$

  Then, $C_1=0$:

$$T_m(r)=T_m(R_m)+\frac{\acute{q}_m}{6k_{eff}}\left(R_m^2-r^2\right), r\in[0,R_m]$$

- **Case 2**: $R_c>0$
  Boundary condition 2 is continuity of the flux (unheated core).

$$-k_{eff}\left(\frac{dT_m}{dr}\right)(R_{core})=0$$

Then, $C_1=\frac{R_{core}^3}{3k_{eff}}$:

$$T_m(r)=T_m(R_m)+\frac{\acute{q}_m}{6k_{eff}}\left(R_m^2-r^2\right)+\frac{q_m R_c^3}{3k_{eff}}\left(\frac{1}{R_m}-\frac{1}{r}\right)$$

Average value

$$\acute{T}_m=\frac{\int_{R_c}^{R_m} T_m(r)r^2\,dr}{\int_{R_c}^{R_m} r^2\,dr}$$

$$\acute{T}_m=T_m(R_m)+\frac{\acute{q}_m}{6k_{eff}}\left(R_m^2-\frac{3}{5}\times\frac{R_m^5-R_c^5}{R_m^3-R_c^3}\right)+q_m\frac{R_c^3}{3k_{eff}}\left(\frac{1}{R_m}-\frac{2}{3}\frac{R_m^3-R_c^3}{R_m^2-R_c^2}\right)$$

This can be rewritten as :

$$\acute{T}_m=T_m(R_m)+\frac{\acute{Q}_p}{V_p}\frac{1}{6k_{eff}}\left(R_m^2-\frac{3}{5}\times\frac{R_m^5-R_c^5}{R_m^3-R_c^3}\right)+\frac{\acute{Q}_p}{V_p}\frac{R_c^3}{3k_{eff}}\left(\frac{1}{R_m}-\frac{2}{3}\frac{R_m^3-R_c^3}{R_m^2-R_c^2}\right)$$

and again:

$$\acute{Q}_p=H_{sTo\,av\,M}\left(T_{m,AV}-T_m\right)$$

where

$$H_{sTo\,av\,M}=\left(\frac{1}{V_p}\frac{1}{6k_{eff}}\left(R_m^2-\frac{3}{5}\times\frac{R_m^5-R_c^5}{R_m^3-R_c^3}\right)+\frac{1}{V_p}\frac{R_c^3}{3k_{eff}}\left(\frac{1}{R_m}-\frac{2}{3}\frac{R_m^3-R_c^3}{R_m^2-R_c^2}\right)\right)^{-1}$$

This is the heat conductance between matrix-shell interface and average matrix temperature.

3. Fuel

- TRISO particle: 5 layers (inner to outer): fuel kernel, graphite buffer, inner PyC, SiC, outer PyC (subscripts f, b, Pin, SiC, Pout, respectively).
- Surface temperature of the TRISO is matrix temperature at which TRISO is (r_p):

$$T_{T,S}=T_m(r_p)$$

- 1D conduction in TRISO particle, power generation only in the fuel kernel.
- Surface temperature of the fuel:

$$T_{f,S}=T_{T,S}+\frac{\acute{Q}_T}{4\pi}\times\left[\frac{1}{k_b}\left(\frac{1}{R_f}-\frac{1}{R_b}\right)+\frac{1}{k_{Pin}}\left(\frac{1}{R_b}-\frac{1}{R_{Pin}}\right)+\frac{1}{k_{SiC}}\left(\frac{1}{R_{Pin}}-\frac{1}{R_{SiC}}\right)+\frac{1}{k_{Pout}}\left(\frac{1}{R_{SiC}}-\frac{1}{R_{Pout}}\right)\right]$$

This can be rewritten as

$$T_{f,S}=T_{T,S}+\frac{\acute{Q}_P}{4\pi N_T}\times\left[\frac{1}{k_b}\left(\frac{1}{R_f}-\frac{1}{R_b}\right)+\frac{1}{k_{Pin}}\left(\frac{1}{R_b}-\frac{1}{R_{Pin}}\right)+\frac{1}{k_{SiC}}\left(\frac{1}{R_{Pin}}-\frac{1}{R_{SiC}}\right)+\frac{1}{k_{Pout}}\left(\frac{1}{R_{SiC}}-\frac{1}{R_{Pout}}\right)\right]$$

and again:

$$\acute{Q}_p=H_{avM\,ToSf}\left(T_{f,S}-T_{m,AV}\right)$$

where

$$H_{avM\,ToSf}=\left(\frac{1}{4\pi N_T}\times\left[\frac{1}{k_b}\left(\frac{1}{R_f}-\frac{1}{R_b}\right)+\frac{1}{k_{Pin}}\left(\frac{1}{R_b}-\frac{1}{R_{Pin}}\right)+\frac{1}{k_{SiC}}\left(\frac{1}{R_{Pin}}-\frac{1}{R_{SiC}}\right)+\frac{1}{k_{Pout}}\left(\frac{1}{R_{SiC}}-\frac{1}{R_{Pout}}\right)\right]\right)^{-1}$$

This is the heat conductance between fuel surface and average matrix temperature. Please notice that we are neglecting the heat resistance between average matrix temperature and surface of the TRISO.

- Profile in the fuel is 1D conduction with heat generation in a sphere:

$$T_f(r)=T_{f,S}+\frac{\acute{q}_f}{6\,k_f}\left(R_f^2-r^2\right)$$

- Maximum fuel temperature: r=0

$$T_{f,max}=T_f(0)=T_f(r)=T_{f,S}+\frac{\acute{q}_f R_f^2}{6\,k_f}$$

- Average fuel temperature:

$$\acute{T}_f=\frac{\int_0^{R_f} T_f(r)\,r^2\,dr}{\int_0^{R_f} r^2\,dr}$$

$$\acute{T}_f=T_{f,S}+\frac{\acute{q}_f R_f^2}{15\,k_f}$$

This can be rewritten as:

$$\acute{T}_f=T_{f,S}+\frac{Q_p}{4\,pi\,R_f^3 N_T}\frac{R_f^2}{15\,k_f}$$

and again:

$$\acute{Q}_p=H\left(T_{f,av}-T_{f,S}\right)$$

where

$$H_{Sf¿}av\,f\,¿=\left(\frac{1}{4\,pi\,R_f^3 N_T}\frac{R_f^2}{15\,k_f}\right)^{-1}\wedge H_{SfTomaxf}=\left(\frac{1}{4\,pi\,R_f^3 N_T}\frac{R_f^2}{6\,k_f}\right)^{-1}$$

The first one is the heat conductance between fuel average and fuel surface. The second one is the heat conductance between the surface of the fuel and the max temperature in the fuel

4. Lumped parameter model

The heat transfer in the pebble and fuel can be modeled as an electric equivalent with 2 nodes:

- 1 node for the graphite average temperature
- 1 node for the fuel average temperature

This requires at least 2 heat conductances

1. The heat conductance between pebble surface and graphite average temperature, which is the harmonic mean of H_shell and H_sToM
2. The heat conductance between graphite average temperature and fuel average temperature, which is the harmonic mean of H_mToSf and H_SmToSf

One can also add a third conductance between fuel average temperature and fuel max temperature. GeN-Foam requires this third resistance, although one can give any value if the max temp is not of interest.

Then, we need two nodes and three heat conductance.
The first one, between the surface of the pebble and the average temperature of the matrix is defined as:

$$H_1 = \frac{H_s H_{sToavM}}{H_s + H_{sToavM}}$$

The second heat conductance makes the link between the previous average temperature of the matrix and the average temperature of the fuel.

$$H_2 = \frac{H_{avMToSf} H_{SfToavf}}{H_{avMToSf} H_{SfToavf}}$$

Finally, we can make the relation between the average fuel temperature and the maximum of the fuel temperature with this third heat conductance:

$$H_3 = \frac{H_{SfToavf} H_{SfTomaxf}}{H_{SfToavf} - H_{SfTomaxf}}$$

We will use a short python script to compute these different values and consider the possible modifications of k or radius values.

Concerning the value of ρ and C_p, we'll use for the first node, the value for graphite and for the second one, the value for fuel. It could be interesting to evaluate the influence of these parameters or take 0 (or something close to 0) as a value of ρC_p to simulate a steady state behavior.

5. Effective conductivity

- Matrix with an effective matrix conductivity k_{eff}, considered homogeneous.
- Packing fraction of TRISO in the matrix:

$$\phi = \frac{N_T V_T}{V_m} = N_T \frac{\frac{4}{3}\pi R_T^3}{\frac{4}{3}\pi\left(R_m^3 - R_c^3\right)} = \frac{N_T R_{Pout}^3}{R_m^3 - R_c^3}$$

- Average conductivity of the TRISO particles:

$$k_T = \frac{\frac{1}{2R_{Pout}}}{\frac{1}{2k_f}\frac{1}{R_f} + \frac{1}{k_b}\left(\frac{1}{R_f} - \frac{1}{R_b}\right) + \frac{1}{k_{Pin}}\left(\frac{1}{R_b} - \frac{1}{R_{Pin}}\right) + \frac{1}{k_{SiC}}\left(\frac{1}{R_{Pin}} - \frac{1}{R_{SiC}}\right) + \frac{1}{k_{Pout}}\left(\frac{1}{R_{SiC}} - \frac{1}{R_{Pout}}\right)}$$

- Maxwell equation with TRISO conductivity k_T, conductivity of the graphite in the matrix k_g and the packing fraction of TRISO in the matrix ϕ. Series of thermal resistance for each layer elements in the TRISO particle.

$$\kappa = \frac{k_T}{k_g}$$

$$\beta = \frac{\kappa - 1}{\kappa + 2}$$

$$k_{eff} = k_g \times \frac{1 + 2\beta\phi}{1 - \beta\phi}$$

6. Interfacial area density and packing fraction

- Interfacial area is defined from contact surface between fluid and structure (subscript st) $A_{f \to st}$ as:

$$IA = \frac{A_{f \to st}}{V_{cell}} = \frac{N_{st} A_{st}}{V_{cell}} = \frac{N_p \times 4\pi R_p^2}{V_{cell}}$$

- Packing fraction of solid in mesh cell α:

$$\alpha = \frac{N_{st} V_{st}}{V_{cell}} = \frac{N_{st} \times \frac{4}{3}\pi R_p^3}{V_{cell}} = \frac{N_{st} \times 4\pi R_p^2}{V_{cell}} \times \frac{R_p}{3} = IA \times \frac{R_p}{3}$$

$$\to IA = \frac{3\alpha}{R_p}$$